Department of Information Technology

University of Gujrat

AI-Powered Web Platform for Custom Upholstery

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DECLARATION

I certify that project title **AL Meeran Upholstery**: **AI-Powered Web Platform for Custom Upholstery** is under my supervision with students of **BS Information Technology (Induction)**, Faculty of Computing & Information Technology, University of Gujrat, Pakistan, worked under my supervision.

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Final Year Project Proposal

1.1 Introduction

This document outlines the proposal for a Final Year Project aimed at developing a modern, feature-rich website for a real-time business solution developed for Al Meeran Upholstery, a local upholstery enterprise in Bahrain, aimed at digitalizing its operations through a modern, responsive e-commerce website integrated with a smart chatbot for customer support, product inquiries, and efficient service management. The primary objective is to create a digital platform that not only showcases the company's services and craftsmanship but also enhances customer engagement and streamlines administrative processes. This proposal defines the project's scope, objectives, system components, architecture, and development plan, providing a clear roadmap for its successful execution.

a. Project Title
b. Project Overview Statement
c. Project Goals
d. Project Objectives
e. High Level System Components
    a. Customer-Facing Website (Frontend)
    b. Admin Panel / CMS (Backend)
    c. AI Chatbot Service
    d. Database
f. List of Optional Functional Units
g. Exclusions
h. Application Architecture
i. Gantt chart

1.2. Project Title:

AL Meeran Upholstery: AI-Powered Web Platform for Custom Upholstery.

1.3. Project Overview statement:

Project Title: **AL Meeran Upholstery:** AI-Powered Web Platform for Custom Upholstery.

Project Manager: Mr. Ikram ul Haq

Project Members:

Name	Registration #	Email Address	Signature
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Project Goal: The primary goal of this Upholstery project is to design and develop a comprehensive, professional, and user-friendly website that serves as a central hub for an upholstery business. The platform aims to attract new clients, streamline the quotation and sample ordering process, and provide an efficient content management system for the business owner.

Objectives:

Sr.#	
1	To develop a visually appealing and fully responsive customer-facing website featuring a gallery, service details, and testimonials.
2	To implement an advanced quote request system allowing users to upload images of their furniture for accurate pricing.
3	To integrate an AI-powered chatbot to provide instant answers to frequently asked questions and guide users.
4	To create a system for customers to browse and order physical fabric samples directly from the website.
5	To build a secure and intuitive Admin Panel for the business owner to manage website content, view quote requests, and process sample orders.
6	To ensure the platform is built on a modern, scalable technology stack to allow for future feature expansion.

Project Success criteria:

Success indicators include a seamless user experience, a fully operational admin panel, and the successful integration of all high-impact features such as the AI chatbot and sample ordering system.

Assumptions, Risks and Obstacles: o **Assumptions:** We assume the availability of necessary hardware, software, and access to APIs for development. o **Risks:** A potential risk is underestimating the complexity of the Admin Panel, which could lead to project delays. o **Obstacles:** Integrating third-party APIs for the chatbot might present technical challenges that require dedicated time for research and troubleshooting.
Organization Address: 407, Al Hadid 104, Bahrain.
Type of project: ❑Research ❑Development
Target End users: Potential customers seeking upholstery services and the owner of Almeeran Upholstery.
Development Technology: ❑Object Oriented ❑Structured
Platform: ❑Web based ❑Distributed ❑Desktop based ❑Setup Configurations ❑Other_____
Approved By: Mr. Usman Sana
Date: 21/October/2025

1.4. Project Goals & Objectives:

Goals:

- To establish a strong, professional online presence for the upholstery business.
- To increase customer lead generation and conversion through an interactive and informative website.
- To improve operational efficiency by automating the initial stages of customer inquiry and quote generation.

Objectives (Measurable):

- Achieve a 100% completion rate for all core features of the customer-facing website and admin panel.
- The AI chatbot must be able to answer at least 80% of pre-defined common customer questions.

- The quote and sample order forms must successfully capture and store user data in the backend database with 100% accuracy.
- The admin panel must allow for the creation, updating, and deletion of gallery items, services, and fabric samples.

1.5. High-level system components:

- **Customer-Facing Website (Frontend):** This is the main interface for users. It will include the homepage, services page, gallery, about us, contact page, quote request form, fabric swatch showcase, and sample ordering system.
- **Admin Panel / CMS (Backend):** A password-protected dashboard for the business owner. This component will allow the admin to manage projects in the gallery, update service descriptions, add/remove fabric samples, view incoming quote requests, and manage sample orders.
- **AI Chatbot Service:** An integrated module that interacts with users in real-time. It will be trained on a set of FAQs to provide instant support.
- **Database:** A central repository to store all application data, including user quote requests, sample order details, gallery content, and fabric information.

1.6. List of optional functional units:

If time permits after the successful implementation of all core components, the following optional features may be developed:

- **User Accounts:** A system for customers to create accounts to track their quote requests and sample orders.
- **3D Fabric Visualization:** An interactive module that allows users to view furniture models and dynamically change fabric textures to preview customization options.
- **Admin Dashboard Analytics:** A simple visual dashboard for the admin to see key metrics, such as the number of new quote requests or popular sample orders.
- **Blog/Care Tips Section:** A simple blog where the admin can post articles about furniture care and upholstery trends.

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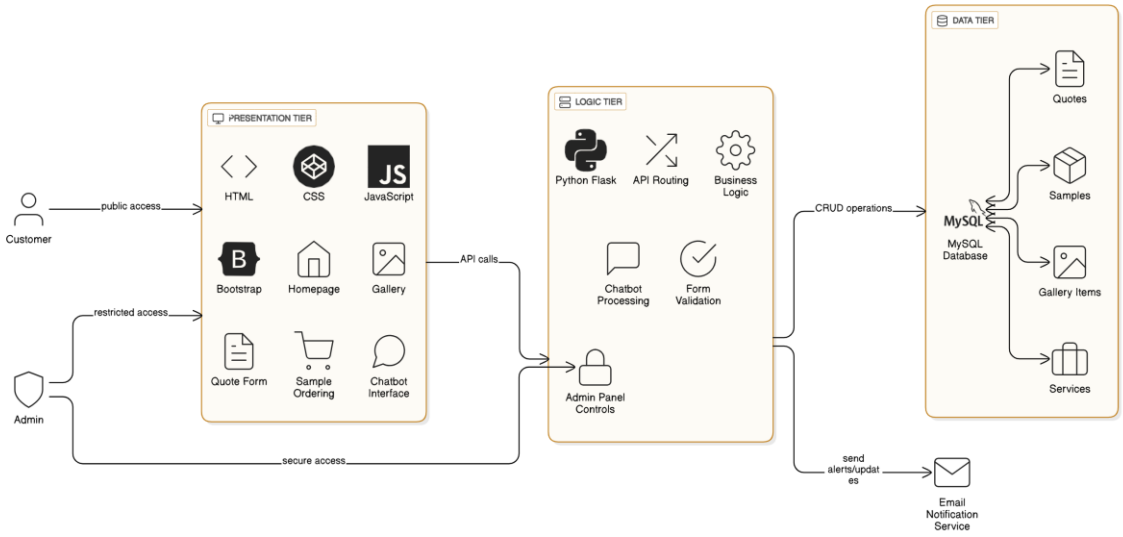
1.7. Exclusions:

The following functionalities are explicitly outside the scope of this project:

- Employee management and scheduling features.
- Full e-commerce or payment processing capabilities.
- Advanced inventory management for fabrics.
- Multi-language support for the website.

1.8. Application Architecture:

This project follows a secure 3-tier architecture with a responsive frontend (HTML, CSS, Bootstrap, JS), a unified Python Flask backend for business logic and AI chatbot processing, and a MySQL database for structured data storage. It supports both customer and admin roles with API-based communication, ensuring efficient data flow and reliable performance.

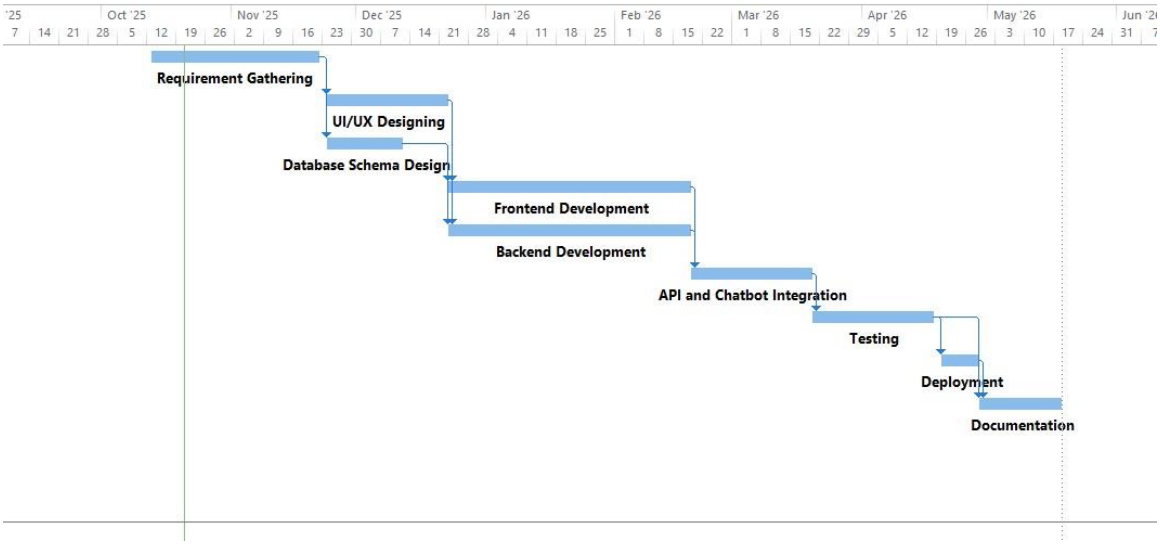


1.9. Gantt chart:

The project is planned to be completed over a 16-week period. The following table outlines the major phases and estimated timeline. Project Months (Oct 2025 – May 2026).

GANTT CHART

	Task Mode	Task Name	Duration	Start	Finish	Predecessors
1		Requirement Gathering	30 days	Mon 10/13/25	Fri 11/21/25	
2		UI/UX Designing	21 days	Mon 11/24/25	Mon 12/22/25	1
3		Database Schema Design	14 days	Mon 11/24/25	Thu 12/11/25	1
4		Frontend Development	42 days	Tue 12/23/25	Wed 2/18/26	2,3
5		Backend Development	42 days	Tue 12/23/25	Wed 2/18/26	2,3
6		API and Chatbot Integration	21 days	Thu 2/19/26	Thu 3/19/26	4,5
7		Testing	21 days	Fri 3/20/26	Fri 4/17/26	6
8		Deployment	7 days	Mon 4/20/26	Tue 4/28/26	7
9		Documentation	14 days	Wed 4/29/26	Mon 5/18/26	7,8



1.10. Hardware and Software Specification:

- **Hardware:** A standard development computer/laptop with at least 8GB RAM and an internet connection.
- **Software:**
  - **Operating System:** Windows
  - **Code Editor:** Visual Studio Code
  - **Server:** XAMPP
  - **Database:** MySQL
  - **AI Framework:** Python Flask
  - **API Testing Tool**: Postman (optional)

1.11. Tools and technologies used with reasoning:

Bootstrap/ HTML/CSS/JS (frontend): To build a responsive and attractive front-end efficiently. Core technologies for web development.

Python (Backend): Python enables fast, clean backend development with easy AI integration.

My SQL(Database): Ensures strong data integrity, stores and retrieves all business, product and customer data.

Python Flask (AI chatbot): Connect the AI chatbot with the main website backend.

XAMPP (Hosting): Provides a local server environment to test and host the project.